

Srivastava implies that a non-diagonalizable pure state on $\mathcal{B}(\ell_2)$ necessarily cannot have multiplicative restriction to any atomic masa (because such restrictions extend to pure states on $\mathcal{B}(\ell_2)$ of the form (D) but the extensions are unique by the positive solution to the Kadison-Singer problem).

Another problem from the paper [9] is whether any pure state on $\mathcal{B}(\ell_2)$ has a multiplicative restriction to some masa of $\mathcal{B}(\ell_2)$. Having multiplicative restriction in this case is equivalent to having the restriction equal to a pure state on the masa. In [1] C. Akemann and N. Weaver provided a negative solution to this problem assuming the continuum hypothesis CH. This, in particular, already showed that Anderson's conjecture is consistently false, but as suggested in [1] it could still be consistent that any pure state on $\mathcal{B}(\ell_2)$ has a multiplicative restriction to a masa. This additional hypothesis in the case of Anderson's conjecture was weakened to MA ([14]) or to $\text{cov}(\mathcal{M}) = \mathfrak{c}$ or to $\mathfrak{d} \leq \mathfrak{p}^*$ (12.5 of [7]), or to another one in [13].

Our counterexample to Anderson's conjecture shows that the additional hypothesis in the result of Akemann and Weaver is not needed when we limit ourself to atomic masas. However, we do not know if our non-diagonalizable pure state can have a multiplicative restriction to a non-atomic masa. An improvement of a result from [9] due to J. Anderson from [2] says that any pure state on a non-atomic masa has many extensions to pure states on $\mathcal{B}(\ell_2)$. So we can say that our pure state is not "determined" by any pure restriction to any masa, i.e., either the restriction is not pure or if it is pure it does not uniquely extends to our pure state.

Our construction is entirely different than that of Akemann and Weaver which used properties of separable C^* -subalgebras of $\mathcal{B}(\ell_2)$ and a well-ordering of all masas in the first uncountable type ω_1 based on the continuum hypothesis to approximate the desired pure state with separable fragments. Let us describe the main idea of our construction here. In a sense, instead of using separable approximations we obtain the desired pure state by approximating it with finite dimensional fragments. Let $\{0, 1\}^m$ denote the set of all sequences of zeros and ones of length $m \in \mathbb{N}$ and let $\{0, 1\}^{\mathbb{N}}$ denote the set of all infinite sequences of zeros and ones. We fix a function $d : \mathbb{N} \rightarrow \mathbb{N}$ which will be specified later and identify ℓ_2 with

$$(I_1) \quad \bigoplus_{m \in \mathbb{N}} \bigotimes_{\sigma \in \{0,1\}^m} \ell_2(d(m)).$$

This can be done by considering a partition of $\mathbb{N}$ into finite sets of sizes $d(m)^{(2^m)}$ for $m \in \mathbb{N}$. Recall that there is a canonical isomorphism

$$(I_2) \quad \mathcal{B}\left(\bigotimes_{\sigma \in \{0,1\}^m} \ell_2(d(m))\right) \cong \bigotimes_{\sigma \in \{0,1\}^m} \mathcal{B}(\ell_2(d(m))).$$

If for every $\sigma \in \{0, 1\}^m$ we choose any non-zero projection $P_\sigma \in \mathcal{B}(\ell_2(d(m)))$ and I denotes the identity, then

$$(P) \quad \prod_{\sigma \in \{0,1\}^m} (I \otimes \dots \otimes \overset{\sigma}{P}_\sigma \otimes \dots \otimes I) = \bigotimes_{\sigma \in \{0,1\}^m} P_\sigma.$$

is a projection that is dominated by any of the projections $(I \otimes \dots \otimes \overset{\sigma}{P}_\sigma \otimes \dots \otimes I)$. So for any $\alpha \in \{0, 1\}^{\mathbb{N}}$ and any choice $v^\alpha = (v_m^\alpha)_{m \in \mathbb{N}} \in \prod_{m \in \mathbb{N}} (\ell_2(d(m)) \setminus \{0\})$ we can define rank one projections $R_{v_m^\alpha} : \ell_2(d(m)) \rightarrow \ell_2(d(m))$ onto the direction of v_m^α and distribute them along the restrictions $\alpha|_m = \alpha|_{\{0, \dots, d(m) - 1\}} \in \{0, 1\}^m$